



Restauration d'images (ou inpainting)

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Sommaire



1. Inpainting ?
2. L'échec MAE
3. Un nouveau départ
4. Adobe 3000

Processus d'inpainting



Jeux de données

CIFAR 10 dans un premier temps (32x32)
pour entrainement et test rapide

- Places365
- CelebA

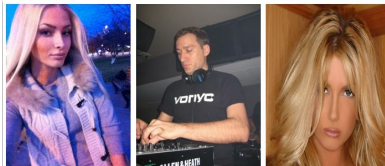
Eyeglasses



Bangs



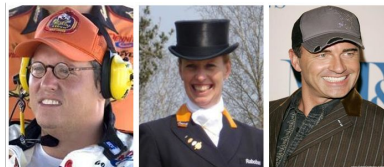
Pointy
Nose



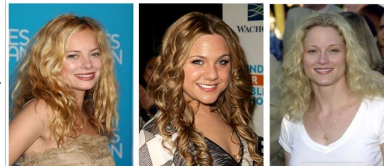
Oval Face



Wearing
Hat



Wavy Hair



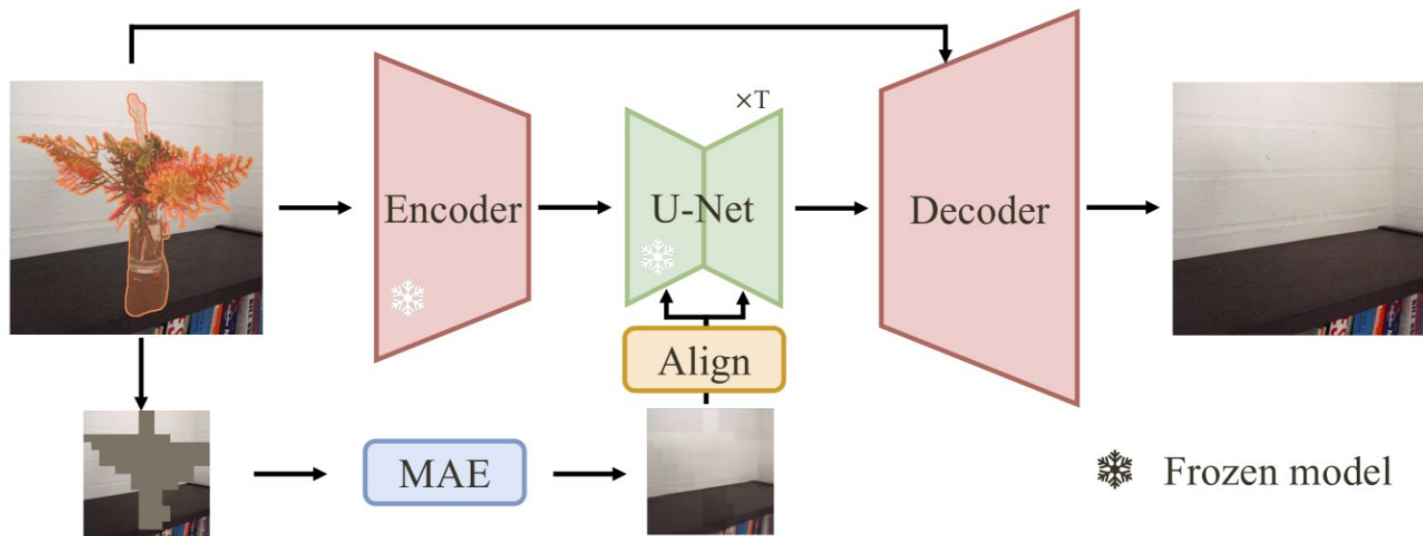
Mustache



Smiling

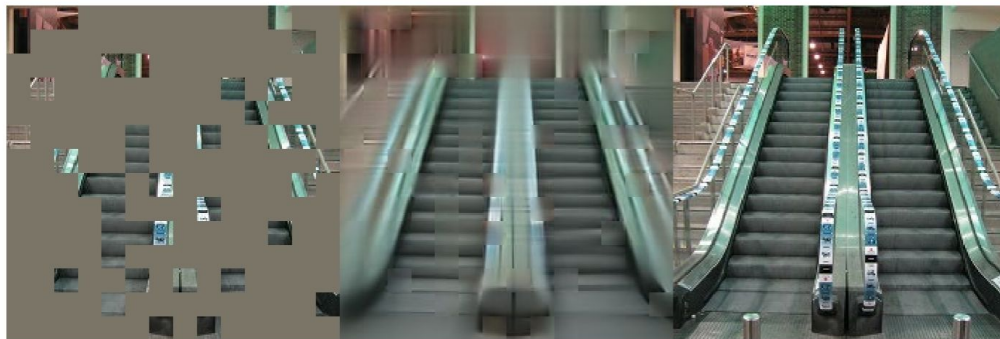


Echec MAE

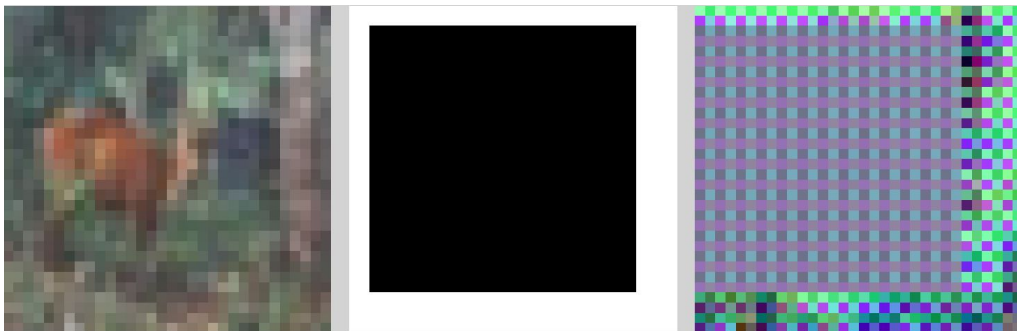


Echec MAE

Attendu:



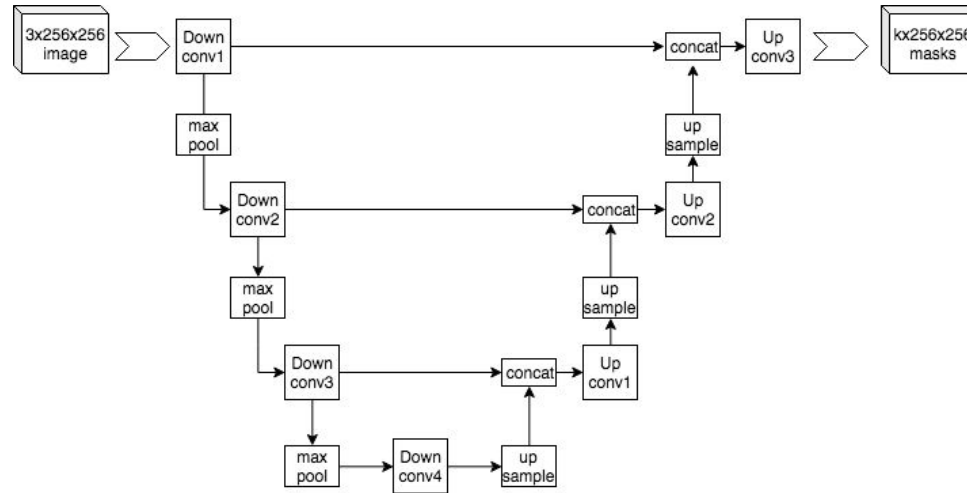
Obtenu:





Nouveau départ : Pen-NET

U-Net



Résultat obtenu avec U-Net

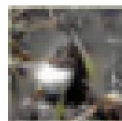
Image originale



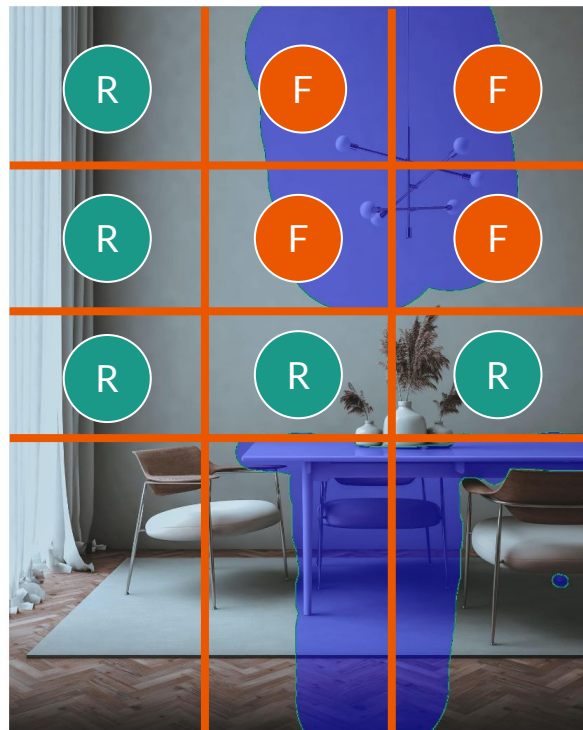
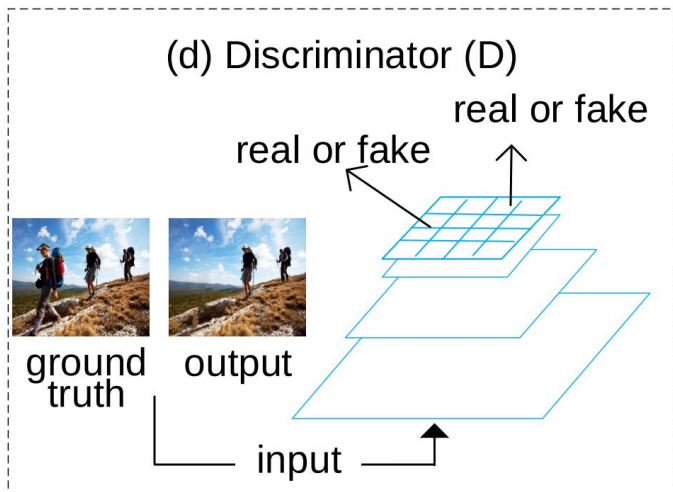
Image masquée



Image générée



Discriminateur



Nouveaux résultats



PSNR de 27.74 dB

Nouveaux résultats

10% de Place2

Image originale



Image masquée

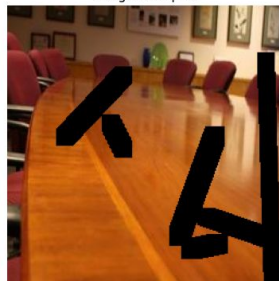


Image générée



Image originale



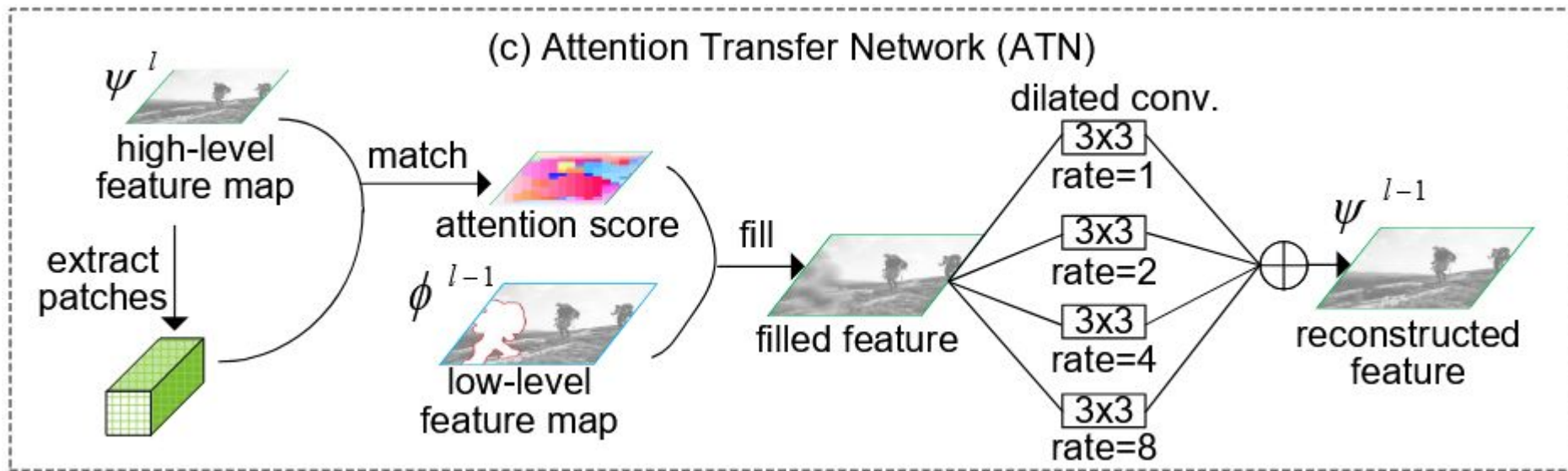
Image masquée



Image générée

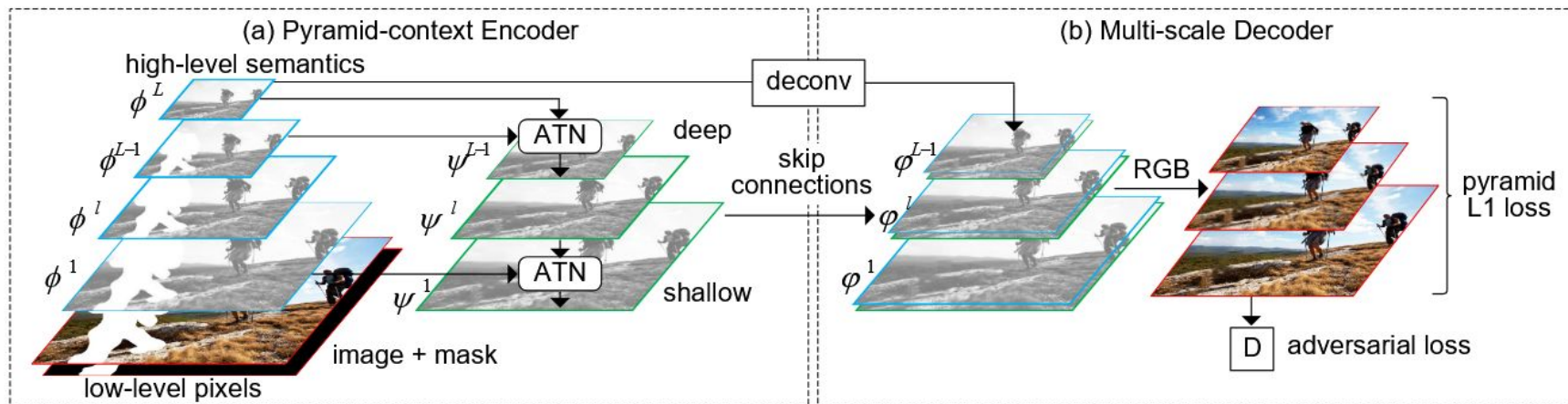


Ajout d'ATN



Permet d'extraire plus de features

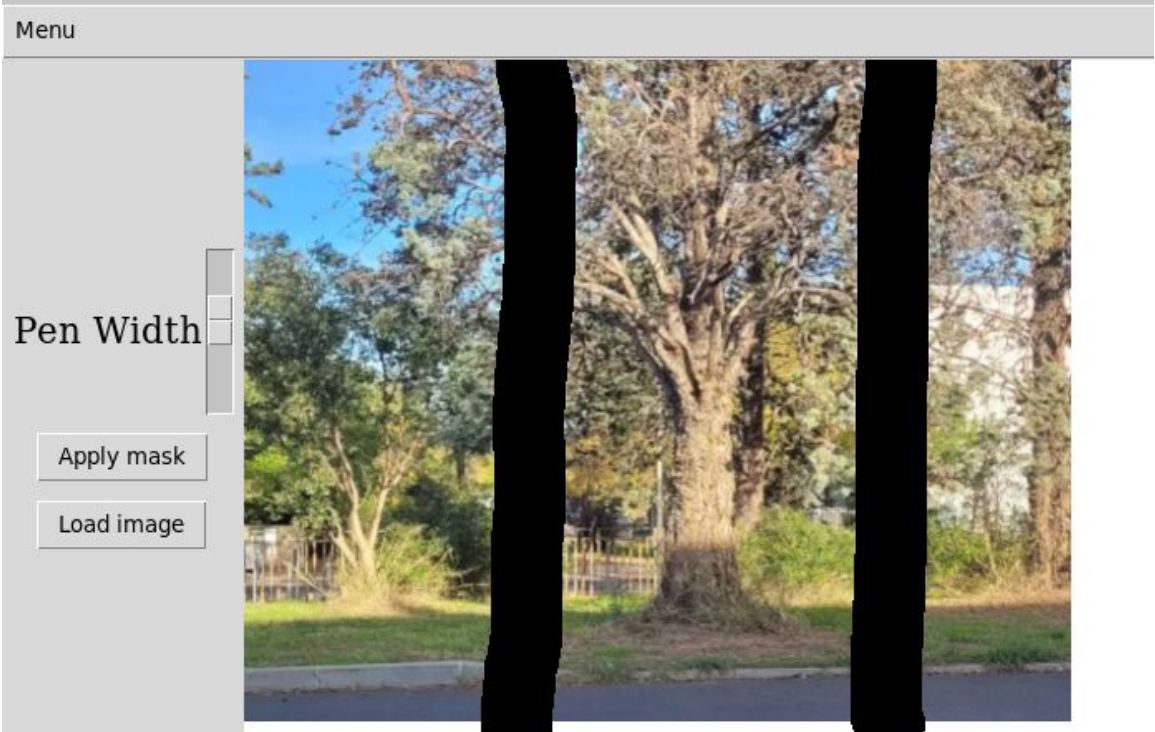
Pen-NET





Adobe 3000

Adobe 3000



Adobe 3000



Image originale

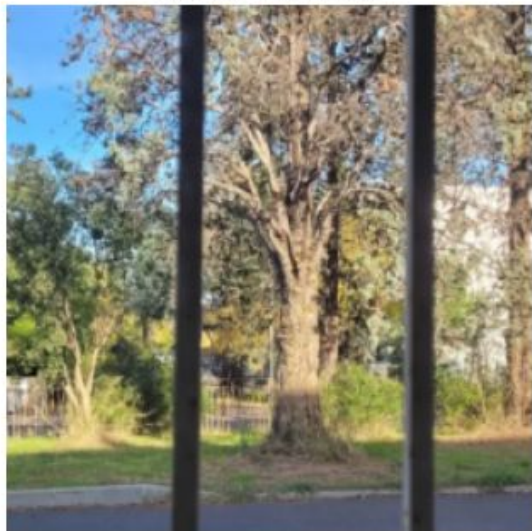
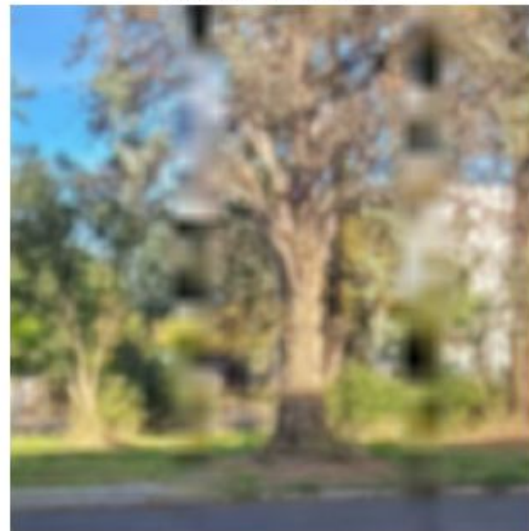


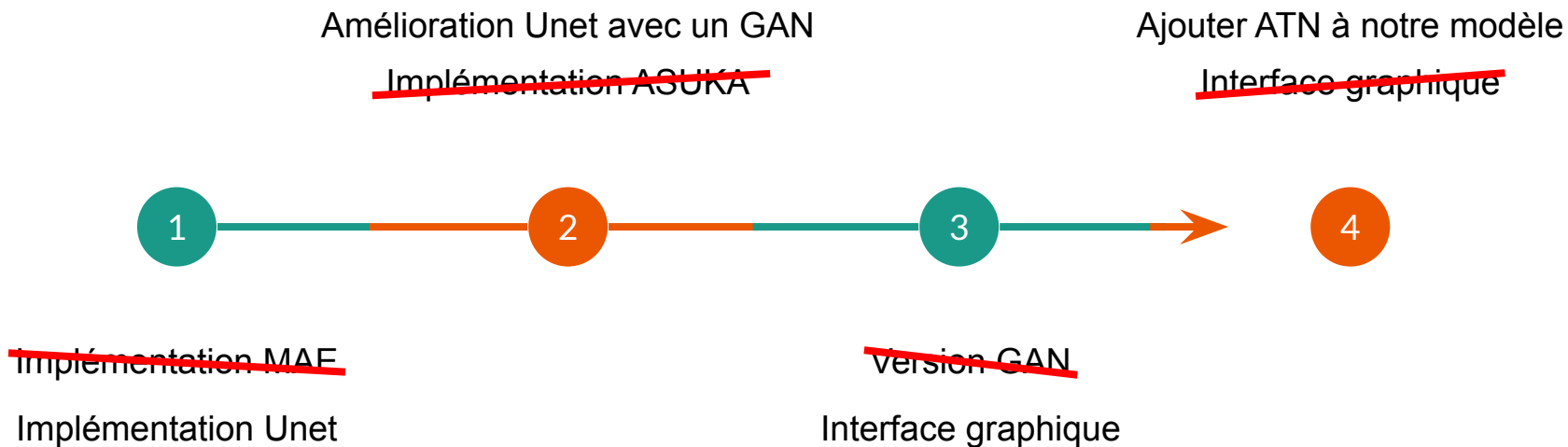
Image masquée



Image générée



Roadmap





Sources

- Learning Pyramid-Context Encoder Network for High-Quality Image Inpainting (2019, cité 656 fois)
<https://arxiv.org/pdf/1904.07475>
- Resolution-robust Large Mask Inpainting with Fourier Convolutions (2021, cité 1409 fois)
<https://arxiv.org/abs/2109.07161>